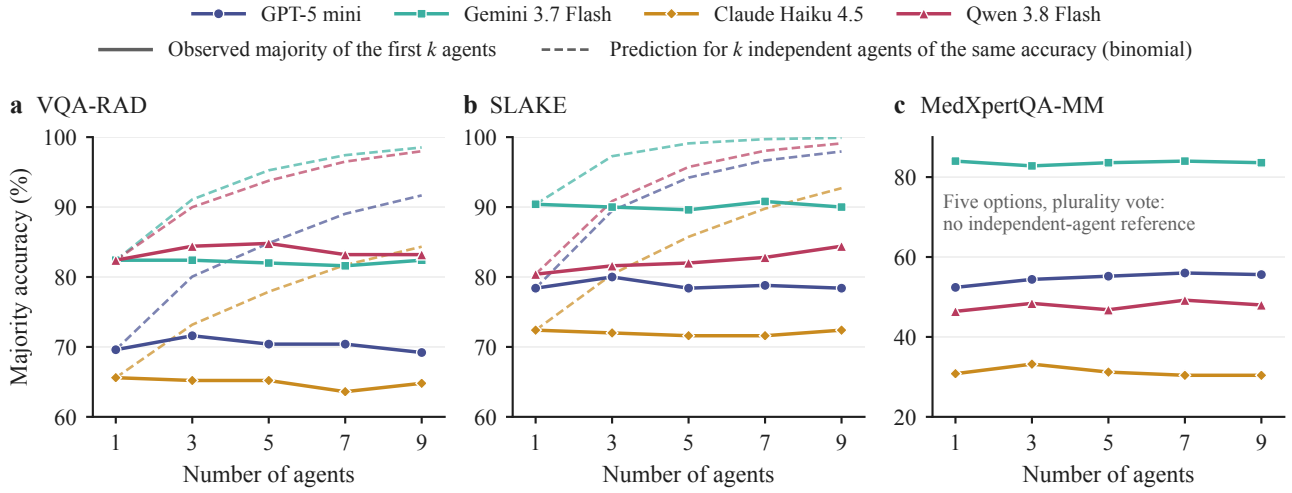
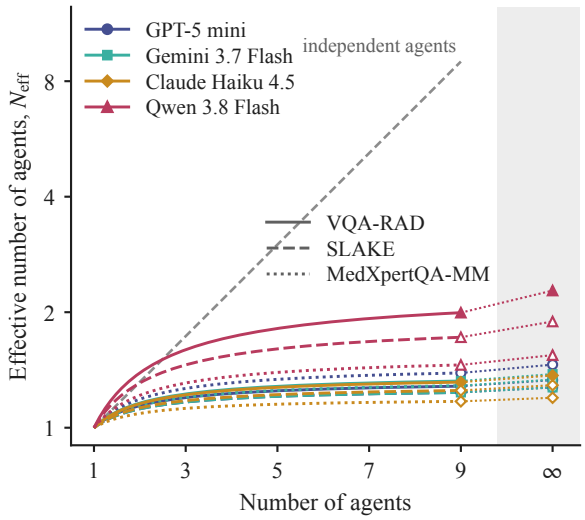
Round 3 candidates: NMI-style displays (figures at print size)

Every figure is drawn at its final CVPR width (column 83.34 mm, text 174.6 mm) and passed the nature-figure QA (alignment, text size, collisions). Centralized, Hybrid and the capability-ladder models (GPT-5 nano, GPT-5, Gemini 3.5 Flash-Lite, Claude Sonnet 5, Qwen 3.8 Max) are still running; the scripts add them when the data land. Tables follow on the last pages.

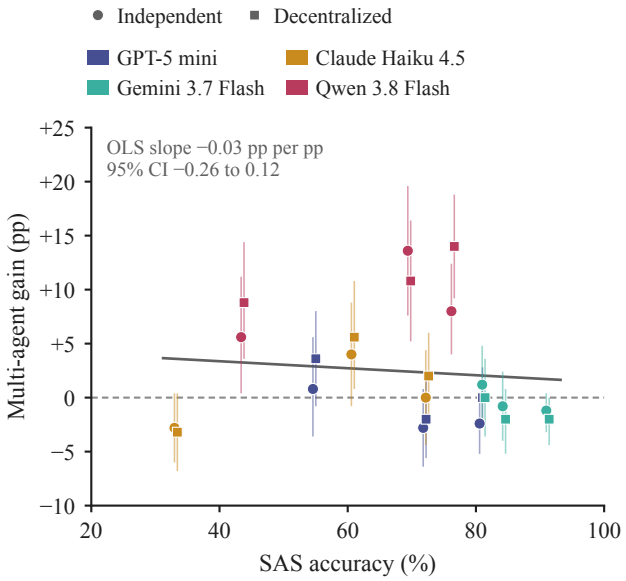
Slot 1: agent scaling (number of agents)



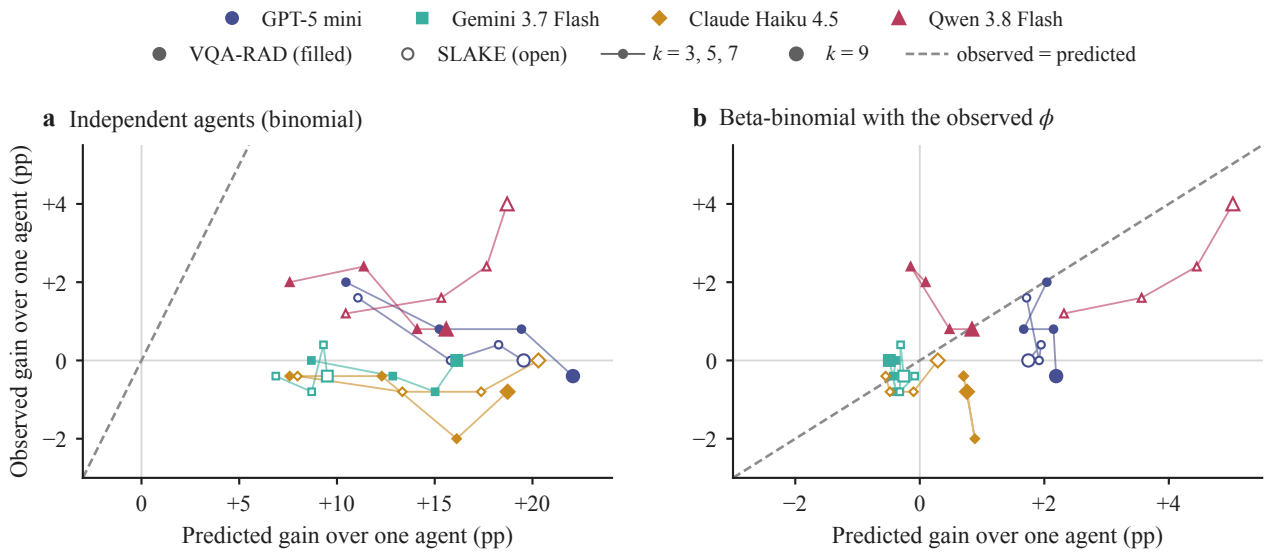
S1a Majority accuracy of the first k agents (solid) against k independent agents of the same accuracy (dashed), per data set.



S1b Effective number of agents saturates near $1/\phi$; dashed, independent agents.

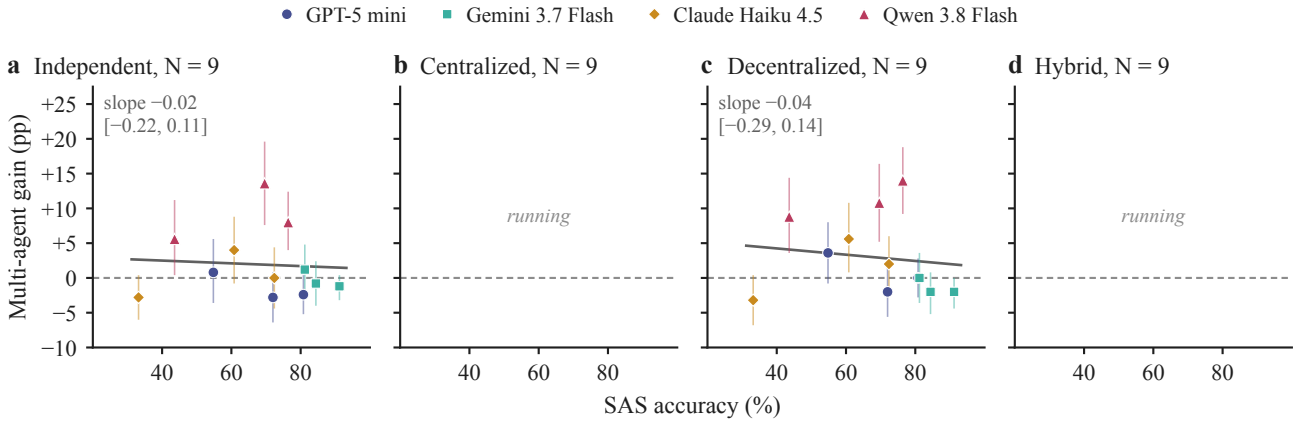


S2a (slot 2) Multi-agent gain against SAS accuracy, marker = architecture, OLS fit.



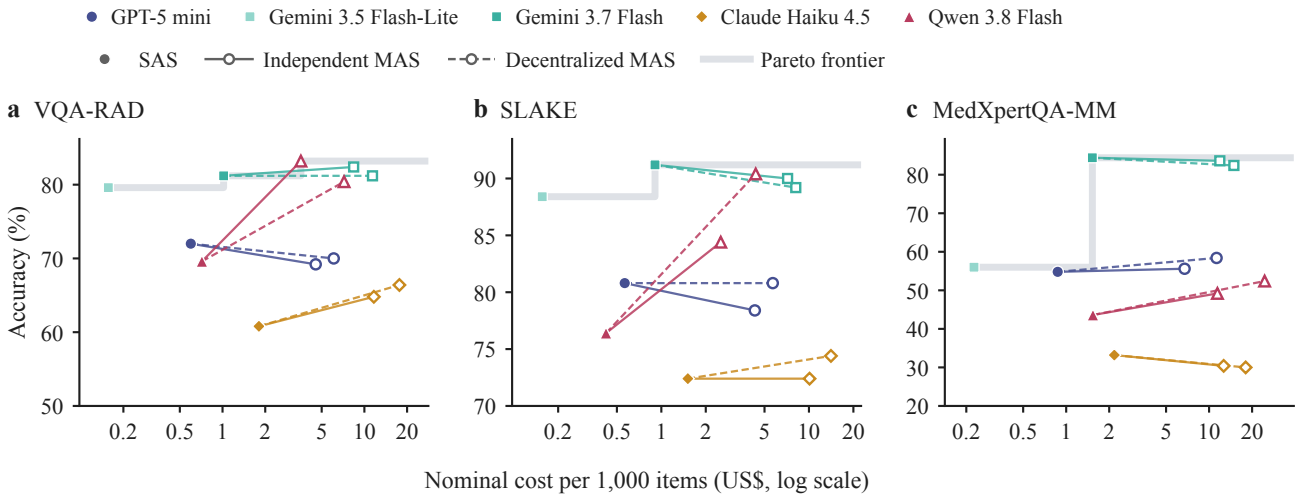
S1c Observed against predicted gain over one agent: independence (a) and beta-binomial with the observed ϕ (b).

Slot 2: capability and multi-agent gain (NMI capability ceiling)

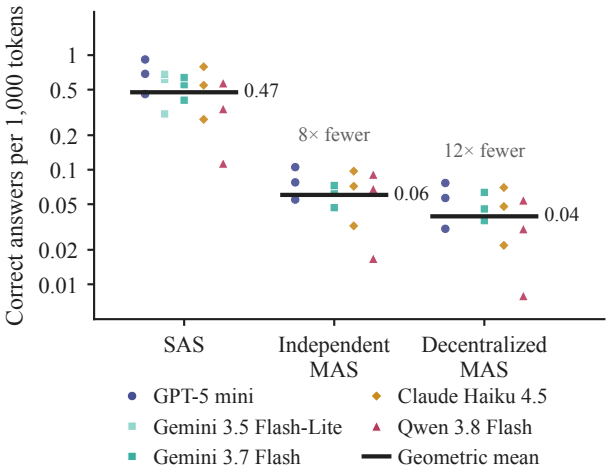


S2b Gain against SAS accuracy, one panel per architecture (Centralized and Hybrid running).

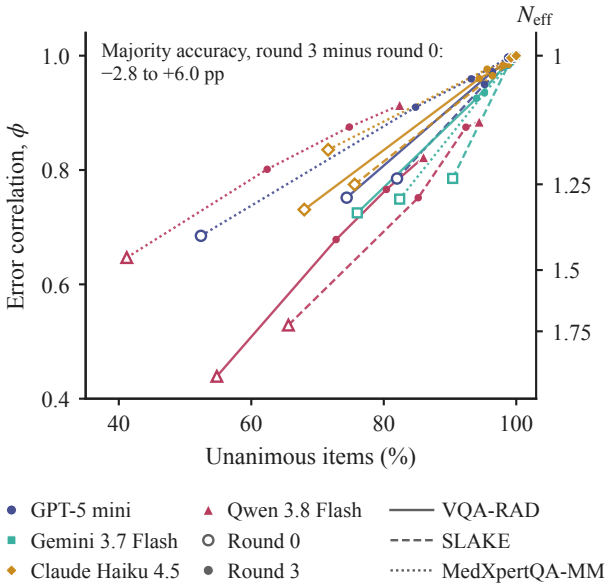
Slot 3: cost and accuracy (NMI economic efficiency)



S3a Accuracy against nominal cost per 1,000 items (log): one fan per model from SAS to each MAS; grey, Pareto frontier.

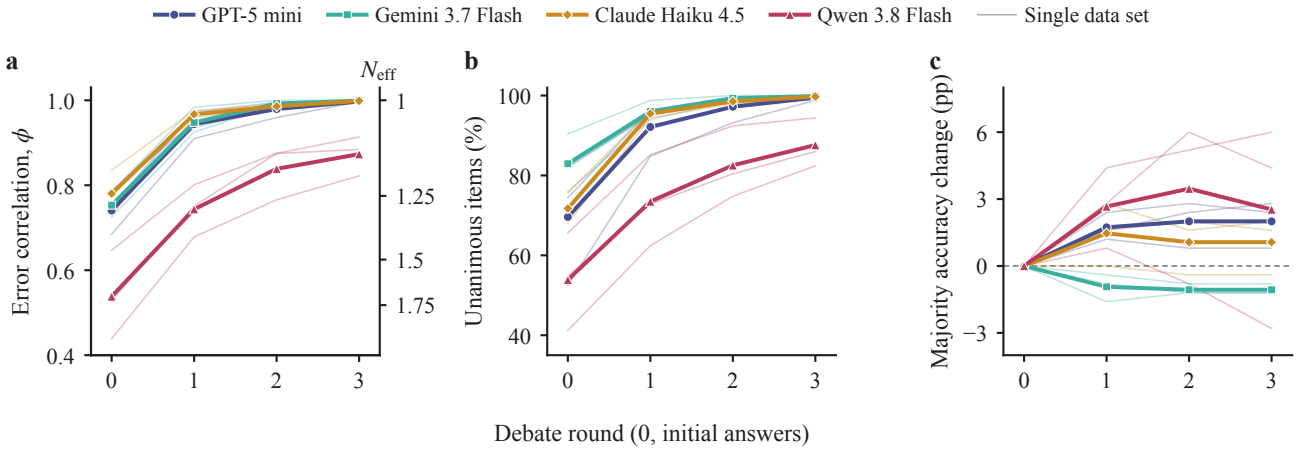


S3b Correct answers per 1,000 tokens by architecture; bar, geometric mean.



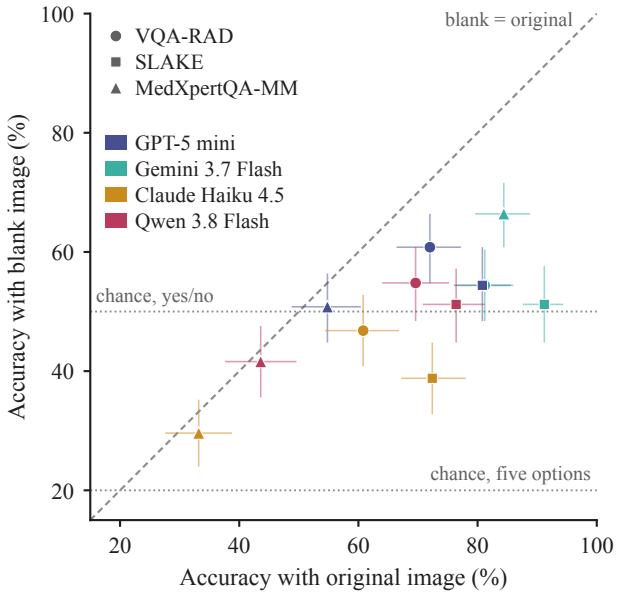
S4d (slot 4) Debate trajectories: unanimity against ϕ from round 0 (open) to round 3 (filled).

Slot 4: debate dynamics (replaces the conformity forest plot)

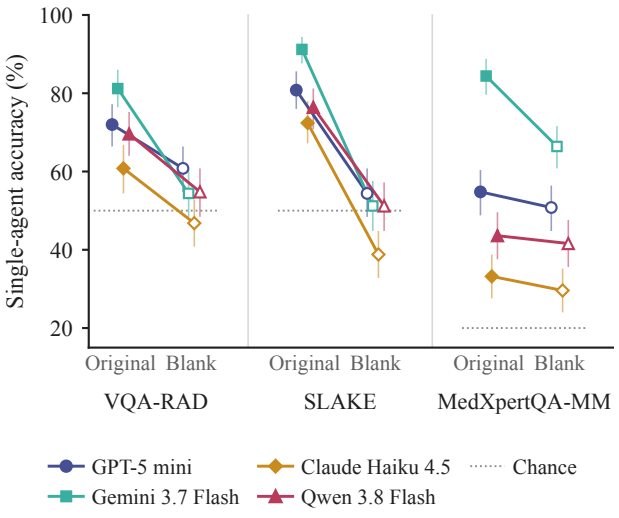


S4c Rounds 0–3 of the Decentralized MAS: ϕ, unanimous items, majority accuracy change.

Slot 5: language prior (replaces the forest panels)



S5c Accuracy with the blank image against the original image; diagonal, no use of the image.



S5d Slope chart from original to blank image per data set.

Table candidates (T1a, T1b, T1b-wide, T1c, T2a, T2b)

Generated from the results files; – marks runs in progress. Pages follow.

[T1a] Main results: accuracy of the five architectures across the capability ladder

Table 1: **Accuracy of the single-agent system and four multi-agent architectures across the capability ladder.** Accuracy (%) of the single-agent system (SAS, shaded) and of the Independent (Ind.), Centralized (Cen.), Decentralized (Dec.) and Hybrid (Hyb.) MAS of 9 agents with original images (250 items per data set). Bold: best architecture per model and data set. Models are grouped by family in order of capability. Item-bootstrap 95% CIs are in the supplement; – marks runs in progress.

		VQA-RAD					SLAKE					MedXpertQA-MM				
Family	Model	SAS	Ind.	Cen.	Dec.	Hyb.	SAS	Ind.	Cen.	Dec.	Hyb.	SAS	Ind.	Cen.	Dec.	Hyb.
OpenAI	GPT-5 nano	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	GPT-5 mini	**72.0**	69.2	–	70.0	–	**80.8**	78.4	–	**80.8**	–	54.8	55.6	–	**58.4**	–
	GPT-5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Google	Gemini 3.5 Flash-Lite	79.6	–	–	–	–	88.4	–	–	–	–	56.0	–	–	–	–
	Gemini 3.7 Flash	81.2	**82.4**	–	81.2	–	**91.2**	90.0	–	89.2	–	**84.4**	83.6	–	82.4	–
Anthropic	Claude Haiku 4.5	60.8	64.8	–	**66.4**	–	72.4	72.4	–	**74.4**	–	**33.2**	30.4	–	30.0	–
	Claude Sonnet 5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Qwen	Qwen 3.8 Flash	69.6	**83.2**	–	80.4	–	76.4	84.4	–	**90.4**	–	43.6	49.2	–	**52.4**	–
	Qwen 3.8 Max	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

[T1b] Multi-agent gain over SAS, averaged over the three data sets

Table 2: **Multi-agent gain over the single-agent system, by model and architecture.** Mean accuracy of the single-agent system (SAS, %) and mean gain of each MAS of 9 agents over SAS (pp), averaged over VQA-RAD, SLAKE and MedXpertQA-MM. Superscript: data sets (of 3) whose paired item-bootstrap 95% CI excludes zero in the direction of the mean; green, gain; red, loss; grey, no data set with such an interval. – marks runs in progress.

		Gain over SAS (pp)			
Model	SAS (%)	Ind.	Cen.	Dec.	Hyb.
OpenAI					
GPT-5 nano	–	–	–	–	–
GPT-5 mini	69.2	-1.5^{0}	–	0.5^{0}	–
GPT-5	–	–	–	–	–
Google					
Gemini 3.5 Flash-Lite	74.7	–	–	–	–
Gemini 3.7 Flash	85.6	-0.3^{0}	–	-1.3^{0}	–
Anthropic					
Claude Haiku 4.5	55.5	0.4^{0}	–	1.5^{1}	–
Claude Sonnet 5	–	–	–	–	–
Qwen					
Qwen 3.8 Flash	63.2	9.1^{3}	–	11.2^{3}	–
Qwen 3.8 Max	–	–	–	–	–

[T1b-wide] Multi-agent gain over SAS per data set and on average

Table 3: **Multi-agent gain over the single-agent system per data set.** Accuracy of each MAS of 9 agents minus SAS accuracy (pp) on VQA-RAD (VQA), SLAKE (SLK) and MedXpertQA-MM (MXQ), and the mean over the three (italic). Bold and coloured: paired item-bootstrap 95% CI excludes zero (green, gain; red, loss); grey: CI includes zero. – marks runs in progress.

		Independent				Centralized				Decentralized				Hybrid			
Family	Model	VQA	SLK	MXQ	Mean	VQA	SLK	MXQ	Mean	VQA	SLK	MXQ	Mean	VQA	SLK	MXQ	Mean
OpenAI	GPT-5 nano	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	GPT-5 mini	−2.8	−2.4	0.8	*−1.5*	–	–	–	–	−2.0	0.0	3.6	*0.5*	–	–	–	–
	GPT-5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Google	Gemini 3.5 Flash-Lite	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	Gemini 3.7 Flash	1.2	−1.2	−0.8	*−0.3*	–	–	–	–	0.0	−2.0	−2.0	*−1.3*	–	–	–	–
Anthropic	Claude Haiku 4.5	4.0	0.0	−2.8	*0.4*	–	–	–	–	**5.6**	2.0	−3.2	*1.5*	–	–	–	–
	Claude Sonnet 5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Qwen	Qwen 3.8 Flash	**13.6**	**8.0**	**5.6**	*9.1*	–	–	–	–	**10.8**	**14.0**	**8.8**	*11.2*	–	–	–	–
	Qwen 3.8 Max	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

[T1c] Architecture summary in the style of NMI Table 2

Table 4: **Every multi-agent architecture multiplies the calls and tokens spent per item.** Means over the model × data-set systems that every listed architecture has completed (n): accuracy (%), gain over the single-agent system (pp; below, range over systems), model calls and tokens (thousands, input plus output) per item, accuracy per 1K tokens, and the error correlation ϕ of the answers that are aggregated (Independent: the 9 agents' answers; Decentralized: final answers after debate; n/a: a single final answer). MAS of 9 agents, original images. – marks runs in progress.

Architecture	n	Acc.	Gain (pp)	Calls	Tokens (K)	Acc./1K tokens	ϕ
SAS	12	68.4	ref.	1.0	1.5	52.4	n/a
Independent	12	70.3	1.9 [−2.8 to 13.6]	9.0	12.2	6.7	0.703
Centralized	–	–	–	–	–	–	–
Decentralized	12	71.3	3.0 [−3.2 to 14.0]	13.2	20.4	4.5	0.967
Hybrid	–	–	–	–	–	–	–

[T2a] Matched component contrasts (replaces the dot-ladder figure)

Table 5: **Changing the model lowers error correlation more than any other tested contrast.** Top: error correlation ϕ of 9 self-consistency agents of one model. Bottom: mean change in error correlation ($\Delta\phi$) when two agents do not share the named component, averaged over models (or model pairs), with item-bootstrap 95% CI. n/a: in-context examples were run on VQA-RAD and SLAKE only.

	VQA-RAD	SLAKE	MedXpertQA-MM
Self-consistency ϕ			
GPT-5 mini	0.741 [0.676, 0.792]	0.800 [0.751, 0.847]	0.692 [0.642, 0.739]
Gemini 3.7 Flash	0.776 [0.714, 0.825]	0.828 [0.754, 0.906]	0.811 [0.743, 0.863]
Claude Haiku 4.5	0.878 [0.836, 0.918]	0.882 [0.836, 0.921]	0.915 [0.885, 0.949]
Qwen 3.8 Flash	0.442 [0.378, 0.504]	0.604 [0.530, 0.666]	0.629 [0.570, 0.685]
$\Delta\phi$ when agents do not share			
Role	−0.048 [−0.071, −0.023]	−0.060 [−0.087, −0.036]	−0.033 [−0.051, −0.014]
Contrast adjustment	−0.092 [−0.123, −0.068]	−0.094 [−0.123, −0.066]	−0.045 [−0.062, −0.027]
In-context examples	−0.110 [−0.142, −0.084]	−0.146 [−0.179, −0.117]	n/a
Report	−0.199 [−0.233, −0.164]	−0.204 [−0.247, −0.162]	−0.172 [−0.211, −0.142]
Perception module	−0.311 [−0.362, −0.260]	−0.275 [−0.334, −0.218]	−0.263 [−0.294, −0.227]
Reasoning module	−0.108 [−0.132, −0.083]	−0.167 [−0.222, −0.129]	−0.430 [−0.477, −0.389]
Model	−0.470 [−0.522, −0.421]	−0.473 [−0.544, −0.406]	−0.492 [−0.536, −0.445]

[T2b] Matched component contrasts with an inline bar per cell

Table 6: **Changing the model lowers error correlation more than any other tested contrast.** Top: error correlation ϕ of 9 self-consistency agents of one model. Bottom: mean change in error correlation ($\Delta\phi$) when two agents do not share the named component, averaged over models (or model pairs), with item-bootstrap 95% CI. n/a: in-context examples were run on VQA-RAD and SLAKE only. Bars: $|\phi|$ or $|\Delta\phi|$ on one 0 to 1 scale (grey, ϕ; blue, $\Delta\phi$).

	VQA-RAD	SLAKE	MedXpertQA-MM
Self-consistency ϕ			
GPT-5 mini	0.741 [0.676, 0.792]	0.800 [0.751, 0.847]	0.692 [0.642, 0.739]
Gemini 3.7 Flash	0.776 [0.714, 0.825]	0.828 [0.754, 0.906]	0.811 [0.743, 0.863]
Claude Haiku 4.5	0.878 [0.836, 0.918]	0.882 [0.836, 0.921]	0.915 [0.885, 0.949]
Qwen 3.8 Flash	0.442 [0.378, 0.504]	0.604 [0.530, 0.666]	0.629 [0.570, 0.685]
$\Delta\phi$ when agents do not share			
Role	−0.048 [−0.071, −0.023]	−0.060 [−0.087, −0.036]	−0.033 [−0.051, −0.014]
Contrast adjustment	−0.092 [−0.123, −0.068]	−0.094 [−0.123, −0.066]	−0.045 [−0.062, −0.027]
In-context examples	−0.110 [−0.142, −0.084]	−0.146 [−0.179, −0.117]	n/a
Report	−0.199 [−0.233, −0.164]	−0.204 [−0.247, −0.162]	−0.172 [−0.211, −0.142]
Perception module	−0.311 [−0.362, −0.260]	−0.275 [−0.334, −0.218]	−0.263 [−0.294, −0.227]
Reasoning module	−0.108 [−0.132, −0.083]	−0.167 [−0.222, −0.129]	−0.430 [−0.477, −0.389]
Model	−0.470 [−0.522, −0.421]	−0.473 [−0.544, −0.406]	−0.492 [−0.536, −0.445]